

Phylogenetics - V

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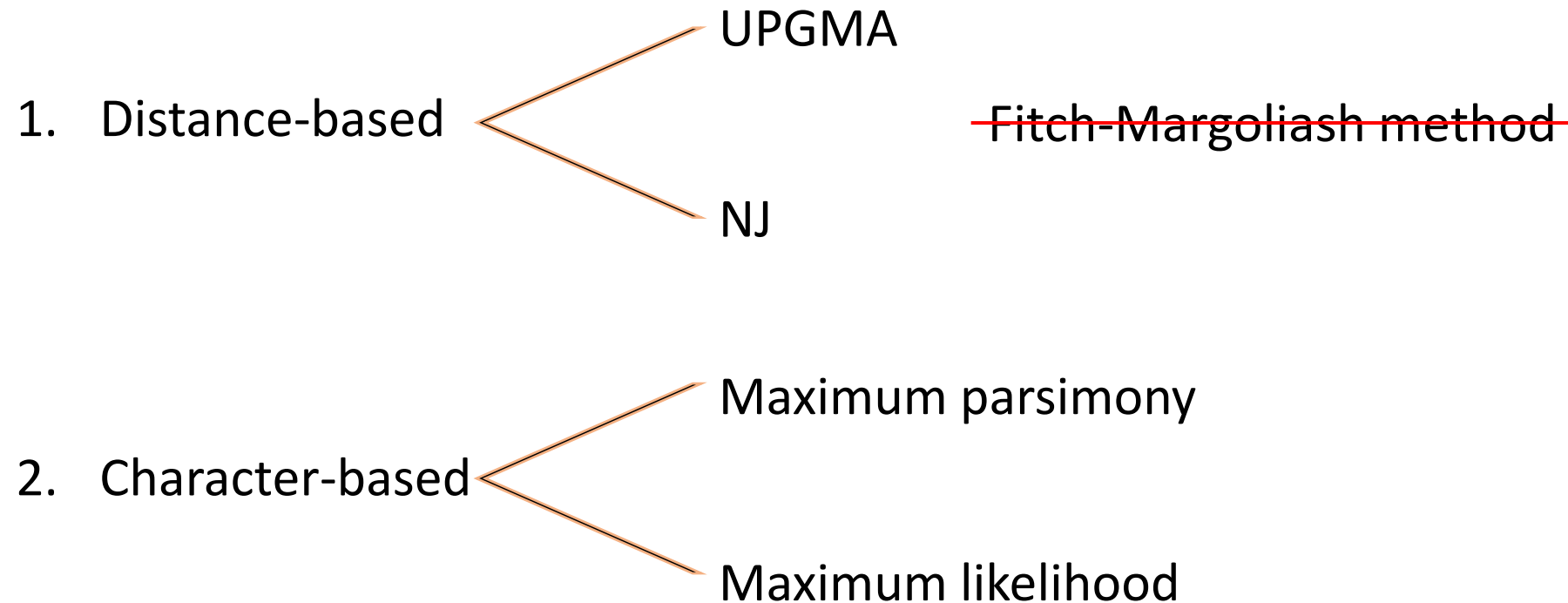
Molecular phylogenetics tree construction can be divided into five steps;

1. Choosing molecular markers
2. Performing Multiple Sequence Alignment (MSA)
3. Choosing a model of evolution
4. Determining a tree building method
5. Accessing tree reliability



Tree building method

Two main methods;



4.2 Character-based methods

taxa \ sites	1	2	3	4	5	6	7	8
		Informative sites			Informative sites		Informative sites	
I	A	A	T	T	A	G	C	T
II	G	G	T	C	G	T	A	G
III	A	A	T	G	C	G	C	T
IV	A	G	T	A	A	G	C	A
V	A	C	T	T	C	G	C	G
VI	A	C	A	T	G	G	C	A

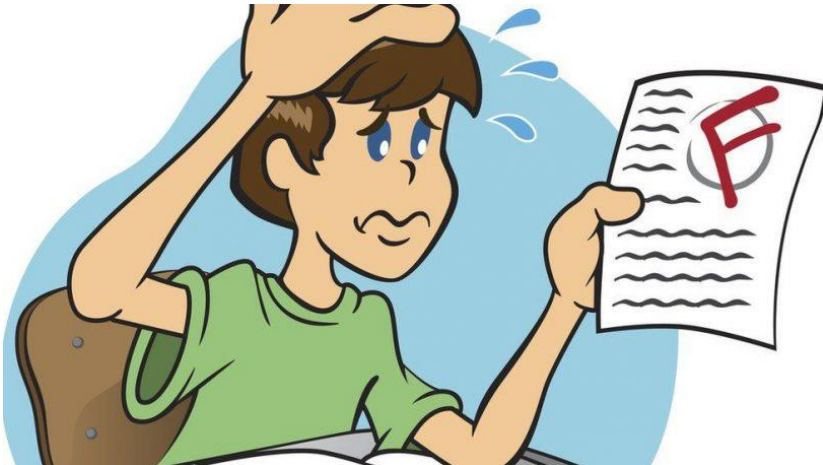
Sites that have at least two different kinds of characters, each occurring at least twice



4.2 Character-based methods

4.2.1 Maximum Parsimony

*The principle of **parsimony** dictates that a theory should provide the simplest possible (viable) explanation for a phenomenon.*

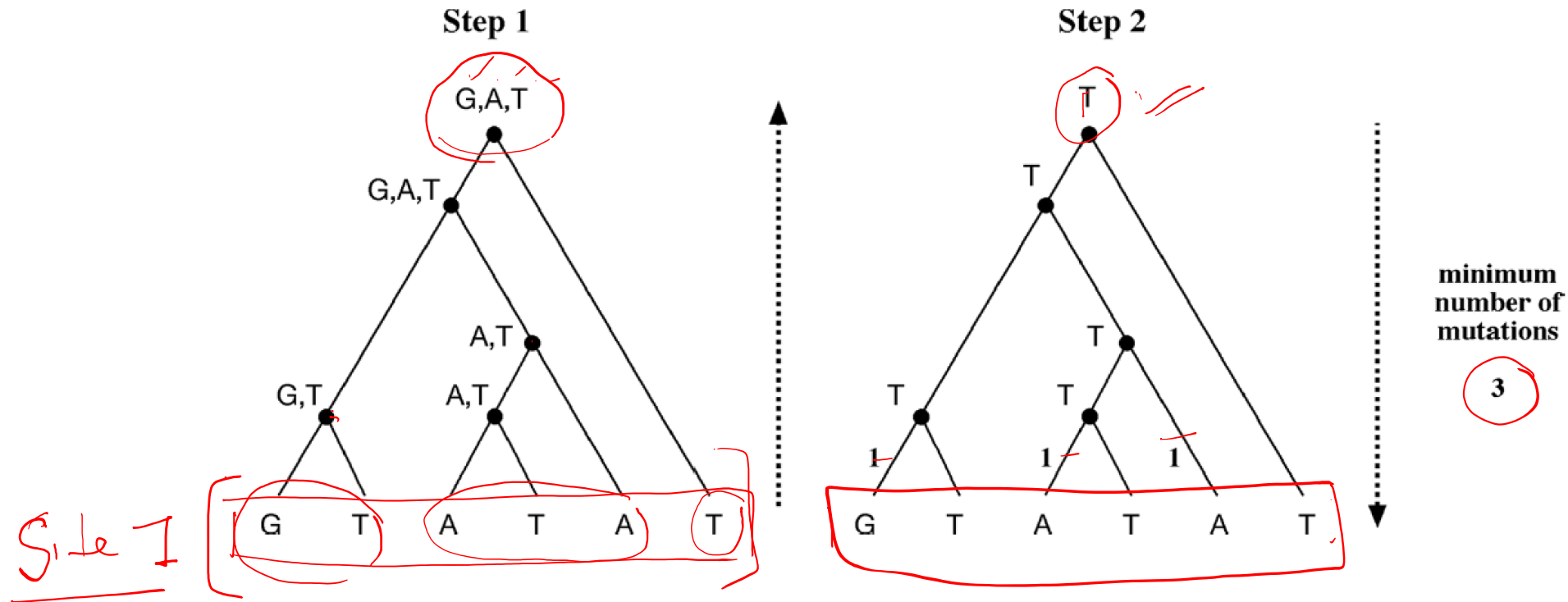


1. Student did not prepare well for the exam.
2. Course instructor did not like the student and awarded an *F*.
3. Mean textbook publisher intentionally wrote wrong tree interpretation in the textbook in hope of sabotaging the career of future bioinformaticians.



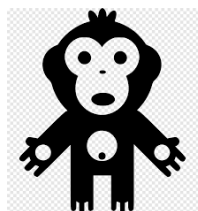
In phylogenetics

The best tree is the one that has fewest evolutionary changes

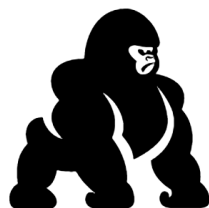




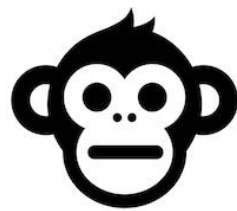
Human



Chimpanzee



Gorilla



Orangutan



T T A G C T A C T



C T A G C T C C C



C T G G C C A C T



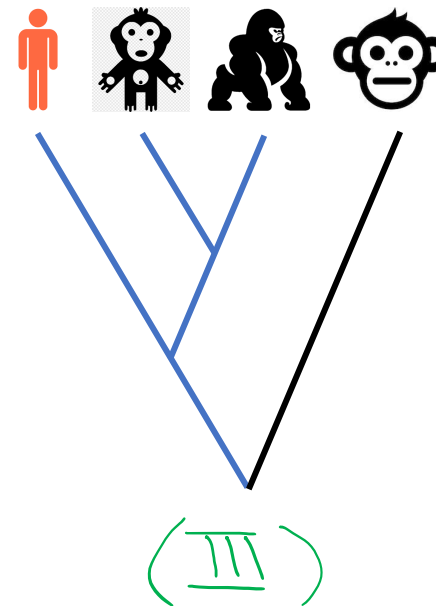
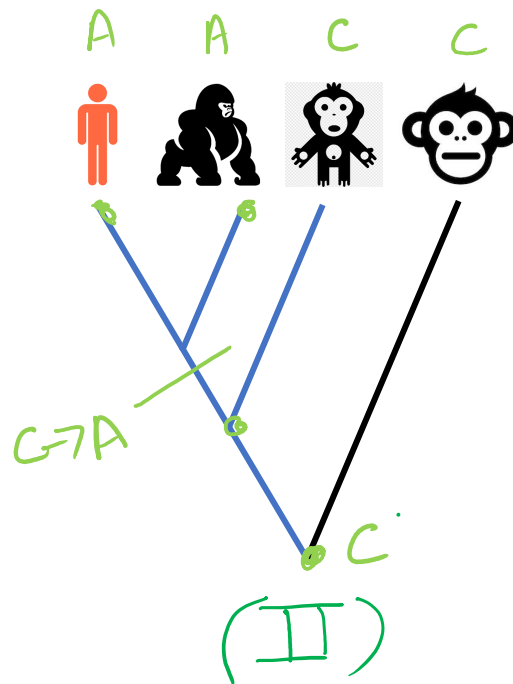
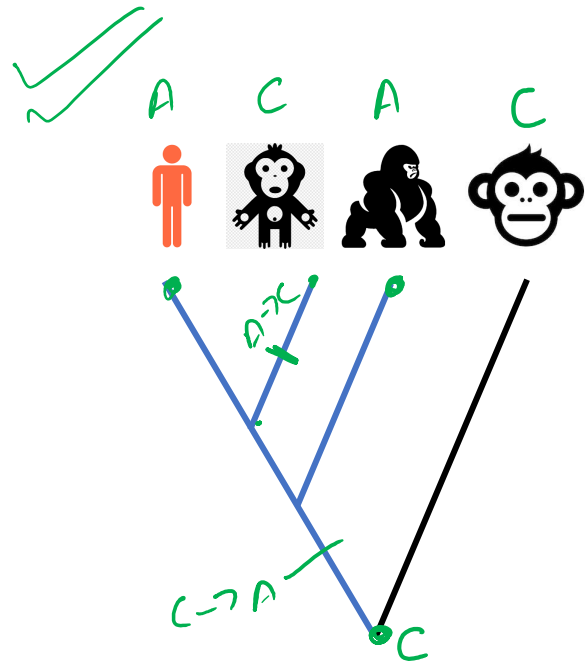
C T G G A C C C T

Gene X

MSA

outgroup





	T	T	A	G	C	T	A	C	T
	C	T	A	G	C	T	C	C	C
	C	T	G	G	C	C	A	C	T
	C	T	G	G	A	C	C	C	T

(MSA)

	Position 3	Position 6	Position 7	Total changes
Tree I	1	1	2	4 ✓
Tree II	2	2	1	5
Tree III	2	2	1	5



4.2 Character-based methods

4.2.2 Maximum Likelihood

- It uses probabilistic models to choose the best tree that has the highest probability.
- It finds the tree that most likely reflects the actual evolutionary process.
- Exhaustive method
 - Searches for every possible tree topology
 - Considers every positions in an alignment
- By employing substitution model, ML calculates the total likelihood of ancestral sequences evolving to internal nodes and eventually to existing sequences.



5. Accessing tree reliability

How reliable is the tree or portion of the tree is?

Given tree is significantly better than another tree?



1. Bootstrapping

Repeatedly sampling trees through slight perturbed datasets

2. Jackknifing

One half of the sites in dataset are randomly deleted, creating datasets half as long as original

